Net Label Mapping on PCB

A. Comms

1. UART0_TX : TXD → PA8 (mcu)
2. UART0_RX : RXD → PA9 (mcu)

3. USB_DP : D+ (J1) → D+ (U2)
4. USB_DM : D- (J1) → D- (U2)

5. On J1 : CC1 → 5.1kΩ → GND
   CC2 → 5.1kΩ → GND

B. CRYSTAL

6. HFXIN : PIN 1 Y1 → PA5. (U1-mcu)
7. HFXOUT : PIN 2 Y1 → PA6 (U1-mcu)

- Note : Y1 and Y2 both grounded through 18pF capacitor

C. SWD HEADER J5 (5 PINS)

Pin 1 : 3.3V
SWDIO : J5 PIN2 → U1 PA19
SWCLK : J5 PIN3 → U1 PA20
RESET : J5 PIN4 → U1 RST_N
Pin 5 : GND

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D LORA : U4 RFM95W

SPI0-CLK : SCK → PA11
SPI0_MOSI : MOSI → PA12
SPI0_MISO : MISO → PA13
LORA_CS : NSS → PA14
LORA_IRQ : DIO0 → PA15
LORA_RST : RESET → PA16

E: BME 680 I2C

I2C_SDA : U5-SDI → PB2
I2C_SCL : U5 SCK → PB3 (4.7k?)
U5-CSB → 3.3V
U5-SDO → GND

Also Note: May have to run 2x 4.7kΩ R's

F: SD CARD SPI → MCU

SPI0 CLK : CLK →
SPI0_MOSI : CMD →
SPI0_MISO : DAT0 →
] shared with LORA
SD_CS : CD/DAT3 → SD_CS (unique) → PA17

F: GPS

J2-PIN 1 → 3V3
J2 PIN 2 → GND
J2 - PIN 3 → UART1_RX : PIN 3 → PA10
J2 - PIN 4 → UART1_TX , PIN 4 → PA 11/26

G: LEDs → MCU GPIO

① LED-PWR: U1 - PB6 → D1
② LED-GPS : U2 - PB7 → D2 } anode
③ LED-LOCA : U1 - PB8 → D3 } gnd: cathode

H SW1 RESET BUTTON

RESET : SW1-PIN1
GND : SW1 - PIN2 →
RESET → R6 one end → 3.3V other end

I. TEST POINTS :

TP1 : 3.3V
TP2 : GND
TP3 : UART0_TX
TP4 : UART0-RX
TP5: SPI0_CLK
TP6: SWDIO

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